PX-400 Firmware 4.8.1 Release Note

Oriole Packaging Systems — Service Engineering

September 2026

1 Applies to

PX-400 systems running firmware 4.8.0 that report intermittent E-217 stops, including Northstar Foods Line 4 (asset NSF-PX400-L4).

2 What changed

- The E-217 diagnostic sampling window is relaxed so brief, self-recovering encoder glitches no longer raise a stop. Genuine signal loss is still detected and still raises E-217.
- Diagnostic D-17 now samples over twenty index cycles instead of ten and reports the worst dropout duration, giving a stronger basis for part decisions.
- No motion-control behavior changes. No parameter migration is required.

3 Field guidance

Keep the existing E-217 procedure: inspect the encoder connector and harness, run D-17, capture the controller log, and replace EA-400 or HN-44 only on diagnostic evidence. Upgrading to 4.8.1 reduces nuisance stops but does not repair a marginal connection, so inspection is still required on assets with repeated E-217 history.

4 Upgrade procedure

1. Confirm the asset is idle and the line is released for maintenance.
2. Back up the controller configuration to the service laptop.
3. Flash firmware 4.8.1 over the service link and verify the version string.
4. Run D-17 once and attach the result to the service work order.

5 Rollback policy

Downgrade to pre-4.8 firmware is not an approved first response to E-217 and requires engineering authorization with a captured fault log. Repeat: inspect first, upgrade second, downgrade never without authorization.

6 Compatibility

Item	Status with 4.8.1
Board PCB-400 Rev B and Rev C	Supported
Diagnostic D-17 (20-cycle)	Supported, replaces 10-cycle D-17
Work orders referencing D-17 results	Remain valid